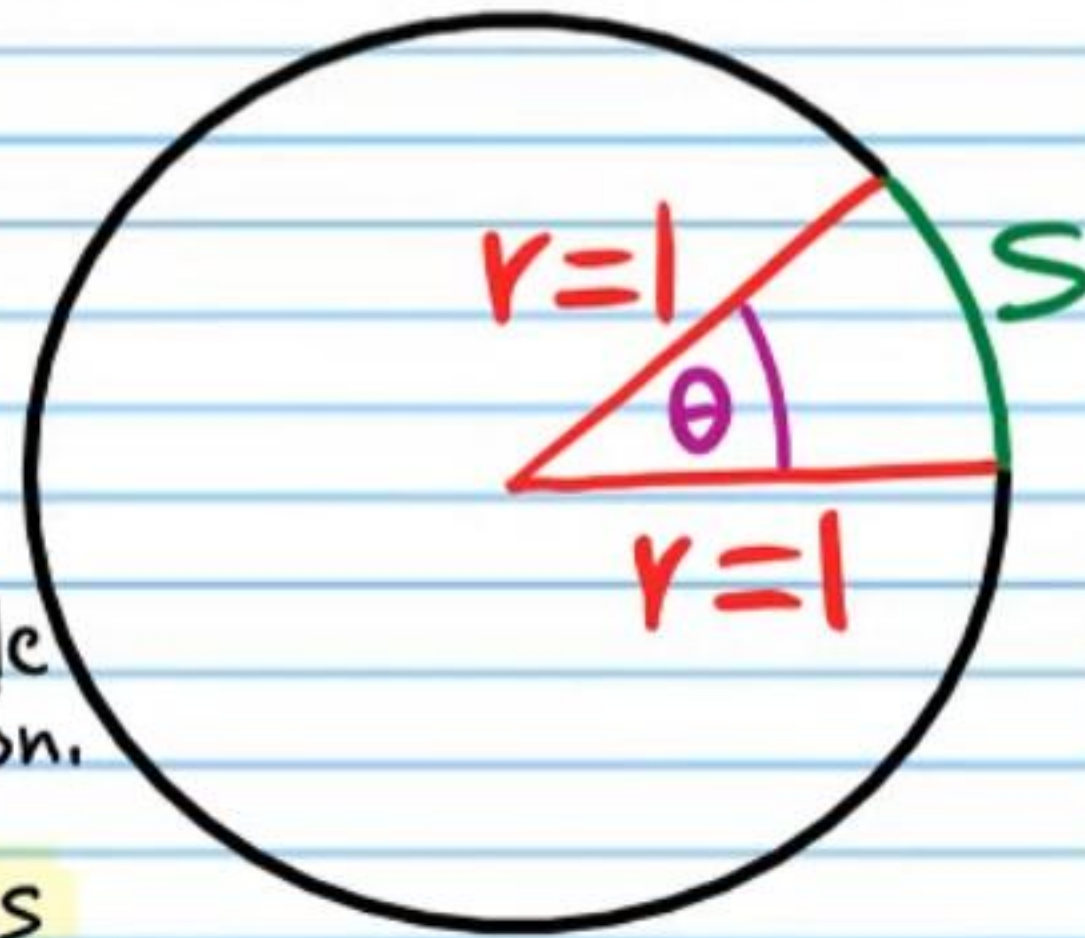


Radians: A Different Unit for the Measure of An Angle

Radians: An alternative unit for the measure of angles and defined to be the length of an arc s on a circle with radius 1 and central angle θ .

Because the circumference of a circle is $2\pi r$, and r is defined to be 1 due to the explanation above, $2\pi(1) = 2\pi$ radians is the angle measure of a complete rotation.



$$2\pi \text{ radians} = 360 \text{ degrees}$$

$$\frac{2\pi}{2} = \frac{360}{2}$$

$$\pi = 180^\circ$$

$$\frac{\pi}{2} = \frac{180}{2}$$

$$\frac{\pi}{2} = 90^\circ$$

$$\frac{\pi}{4} = \frac{180}{4}$$

$$\frac{\pi}{4} = 45^\circ$$

$$\frac{\pi}{6} = \frac{180}{6}$$

$$\frac{\pi}{6} = 30^\circ$$

$$\frac{\pi}{3} = \frac{180}{3}$$

$$\frac{\pi}{3} = 60^\circ$$

How to Convert between Degrees and Radians

The Factor-Label Method

Converting from Degrees to Radians

$$\text{degree measure} \times \frac{\pi \text{ radians}}{180 \text{ degrees}} = \text{radian measure}$$

$$\text{Example: } 36^\circ \times \frac{\pi}{180} = \frac{36\pi}{180} = \boxed{\frac{\pi}{5}}$$

Converting from Radians to Degrees

$$\text{Radian measure} \times \frac{180 \text{ degrees}}{\pi \text{ radians}} = \text{degree measure}$$

$$\text{Example: } \frac{2\pi}{3} \times \frac{180}{\pi} = \frac{360\pi}{3\pi} = \frac{360}{3} = \boxed{120^\circ}$$

Practice

Convert the following angle measures to radian units.

① 80°

$$80^\circ \times \frac{\pi}{180} = \frac{80\pi}{180} = \boxed{\frac{4\pi}{9}}$$

② 150°

$$150^\circ \times \frac{\pi}{180} = \frac{150\pi}{180} = \boxed{\frac{5\pi}{6}}$$

$$\textcircled{3} \ 315^\circ \quad 315^\circ \times \frac{\pi}{180^\circ} = \frac{315\pi}{180} = \boxed{\frac{7\pi}{4}}$$

$$\textcircled{4} \ 1^\circ \quad 1^\circ \times \frac{\pi}{180^\circ} = \boxed{\frac{\pi}{180}}$$

$$\bullet \textcircled{5} \ 700^\circ \quad 700^\circ \times \frac{\pi}{180^\circ} = \frac{700\pi}{180} = \boxed{\frac{35\pi}{9}}$$

$$\textcircled{6} \ 57^\circ \quad 57^\circ \times \frac{\pi}{180^\circ} = \frac{57\pi}{180} = \boxed{\frac{19\pi}{60}}$$

$$\textcircled{7} \ 10^\circ \quad 10^\circ \times \frac{\pi}{180^\circ} = \frac{10\pi}{180} = \boxed{\frac{\pi}{18}}$$

$$\textcircled{8} \ 275^\circ \quad 275^\circ \times \frac{\pi}{180^\circ} = \frac{275\pi}{180} = \boxed{\frac{55\pi}{36}}$$

Convert the following angle measures to degree units.

$$\bullet \textcircled{9} \ \frac{\pi}{12} \quad \frac{\pi}{12} \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{12\pi} = \frac{180^\circ}{12} = \boxed{15^\circ}$$

$$\textcircled{10} \ \frac{3\pi}{4} \quad \frac{3\pi}{4} \times \frac{180^\circ}{\pi} = \frac{540\pi^\circ}{4\pi} = \frac{540^\circ}{4} = \boxed{135^\circ}$$

$$\textcircled{11} \ \frac{7\pi}{6} \quad \frac{7\pi}{6} \times \frac{180^\circ}{\pi} = \frac{1260\pi^\circ}{6\pi} = \boxed{210^\circ}$$

* Round to the nearest thousandth

$$\textcircled{12} \ 8 \quad 8 \times \frac{180^\circ}{\pi} = \frac{1440^\circ}{\pi} \approx \boxed{458.366^\circ}$$

$$\textcircled{13} \ 2.64 \quad 2.64 \times \frac{180^\circ}{\pi} \approx \boxed{151.261^\circ}$$

$$\bullet \textcircled{14} \ \frac{\pi}{10} \quad \frac{\pi}{10} \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{10\pi} = \frac{180^\circ}{10} = \boxed{18^\circ}$$

$$\textcircled{15} \ \frac{5\pi}{4} \quad \frac{5\pi}{4} \times \frac{180^\circ}{\pi} = \frac{900\pi^\circ}{4\pi} = \frac{900^\circ}{4} = \boxed{225^\circ}$$

$$\textcircled{16} \ \frac{7\pi}{3} \quad \frac{7\pi}{3} \times \frac{180^\circ}{\pi} = \frac{1260\pi^\circ}{3\pi} = \boxed{420^\circ}$$

Round to the nearest thousandth

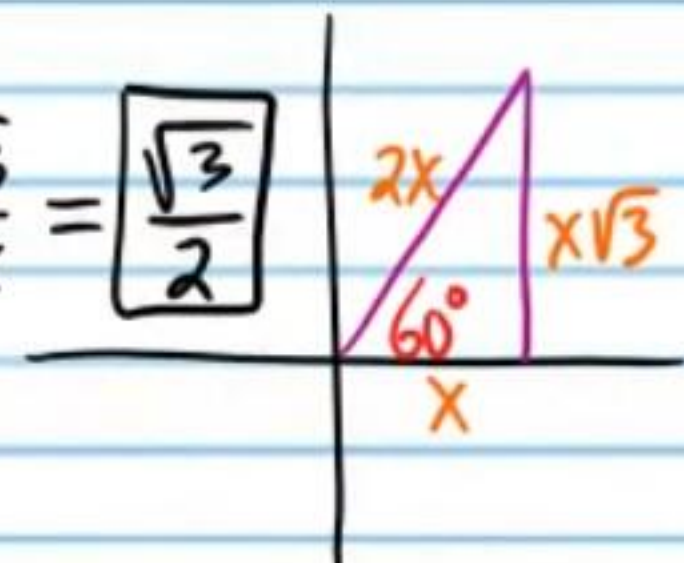
$$\textcircled{17} \ 6 \quad 6 \times \frac{180^\circ}{\pi} \approx \boxed{343.775^\circ}$$

Using Radians with Trigonometry

Find the values of the following expressions without a calculator.

• (18) $\sin \frac{\pi}{3}$

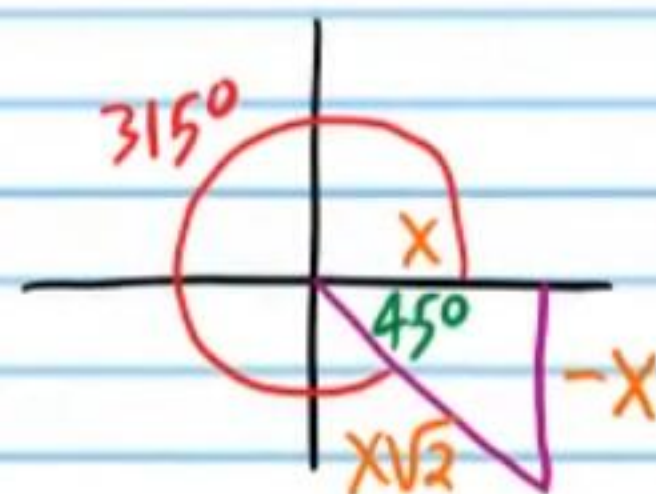
$$\frac{\pi}{3} \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{3\pi} = 60^\circ$$

$$\sin \frac{\pi}{3} = \sin 60^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$


(19) $\sec \frac{7\pi}{4}$

$$\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = \frac{1260\pi^\circ}{4\pi} = 315^\circ$$

$$\sec \frac{7\pi}{4} = \sec 315^\circ = \frac{x\sqrt{2}}{x} = \boxed{\sqrt{2}}$$



(20) $\tan \pi$

$$\pi \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{\pi} = 180^\circ$$

$$\tan \pi = \tan 180^\circ = \boxed{0}$$

(21) $\cos \frac{\pi}{2}$

$$\frac{\pi}{2} \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{2\pi} = \frac{180^\circ}{2} = 90^\circ$$

$$\cos \frac{\pi}{2} = \cos 90^\circ = \boxed{0}$$

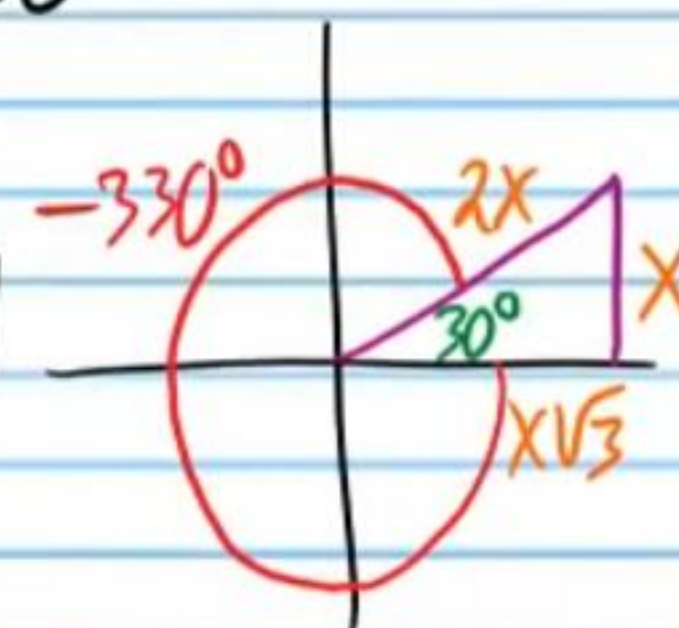
(22) $\csc \frac{3\pi}{2}$ $\frac{3\pi}{2} \times \frac{180^\circ}{\pi} = \frac{540\pi^\circ}{2\pi} = 270^\circ$

$$\csc \frac{3\pi}{2} = \csc 270^\circ = \boxed{-1}$$

(23) $\cot(-\frac{11\pi}{6})$

$$-\frac{11\pi}{6} \times \frac{180^\circ}{\pi} = -\frac{1980\pi^\circ}{6\pi} = -330^\circ$$

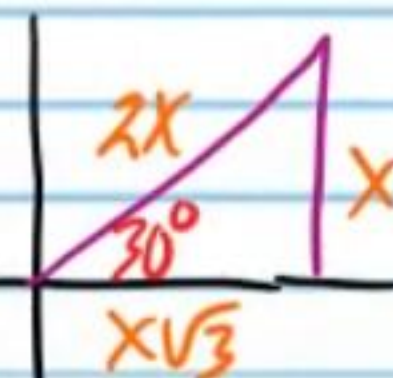
$$\cot(-\frac{11\pi}{6}) = \cot(-330^\circ) = \frac{x\sqrt{3}}{x} = \boxed{\sqrt{3}}$$



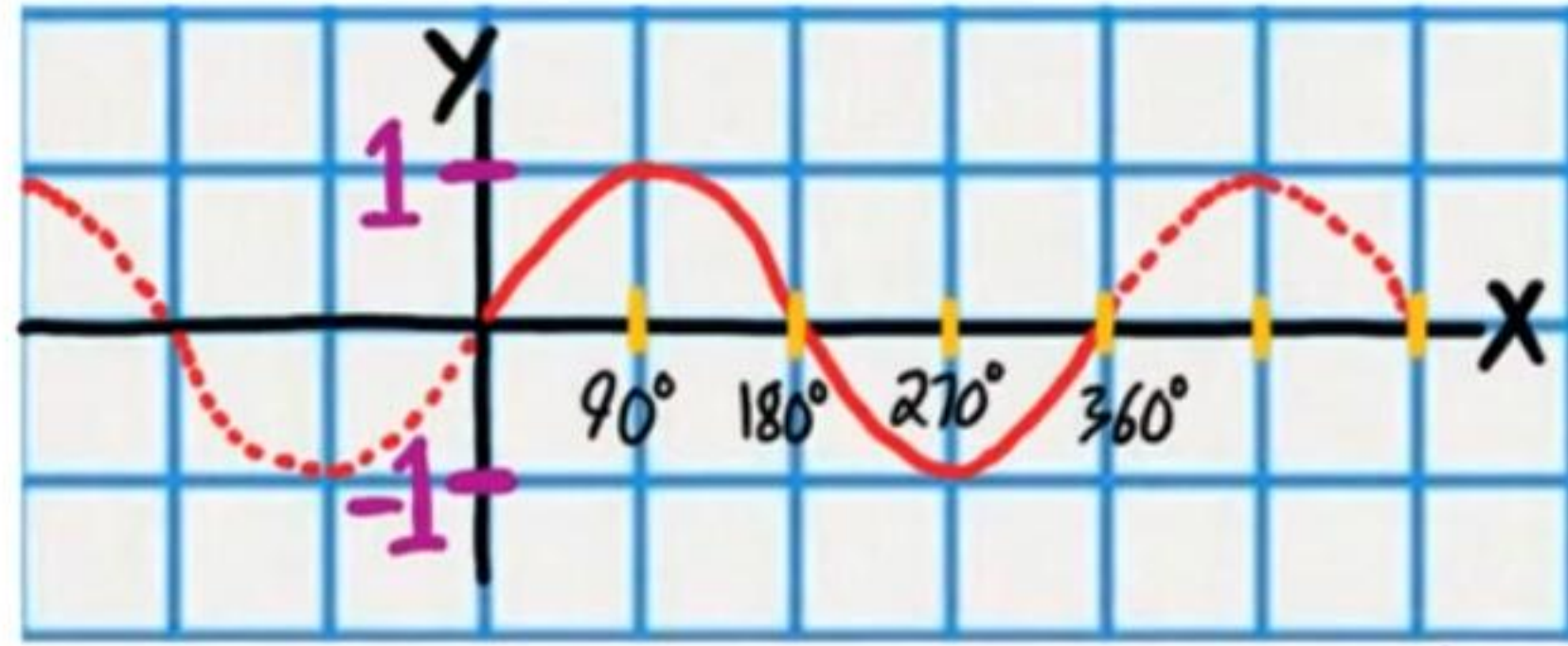
• (24) $\cos \frac{\pi}{6}$

$$\frac{\pi}{6} \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{6\pi} = 30^\circ$$

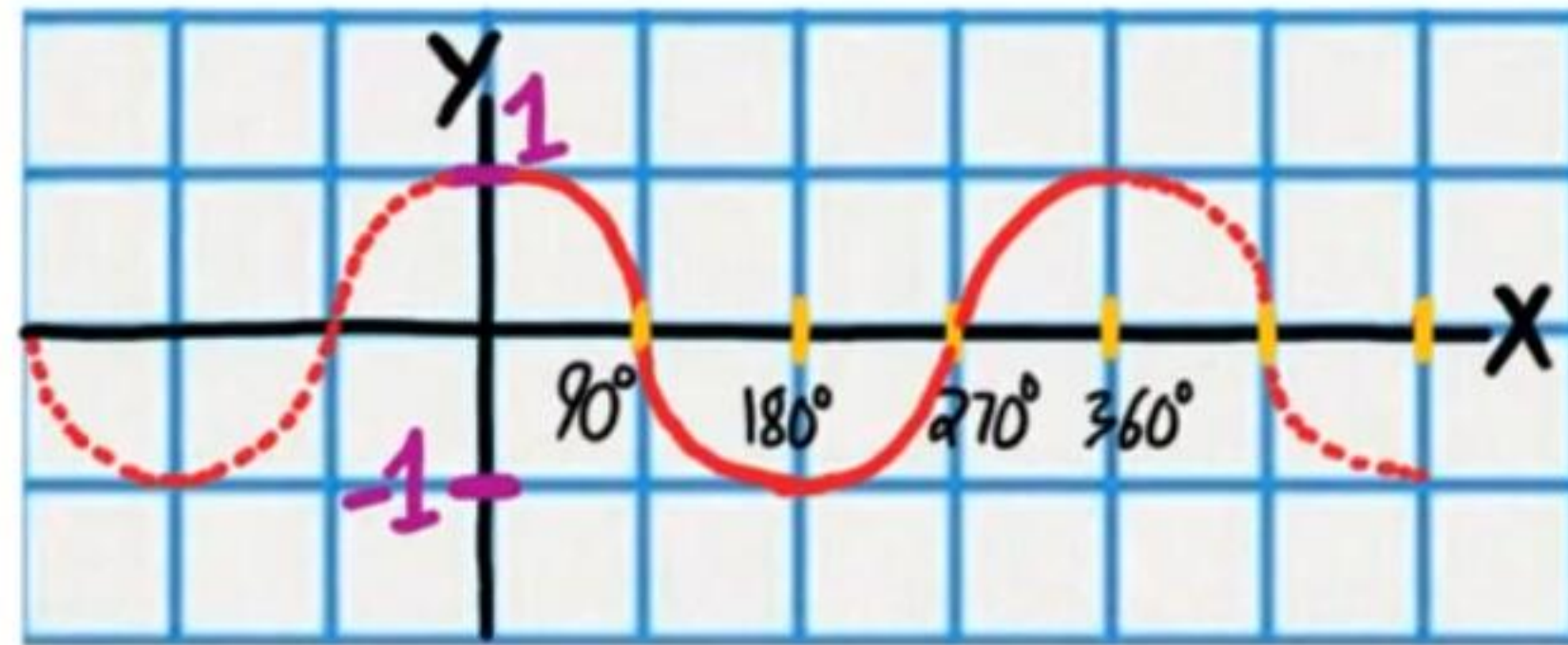
$$\cos \frac{\pi}{6} = \cos 30^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$



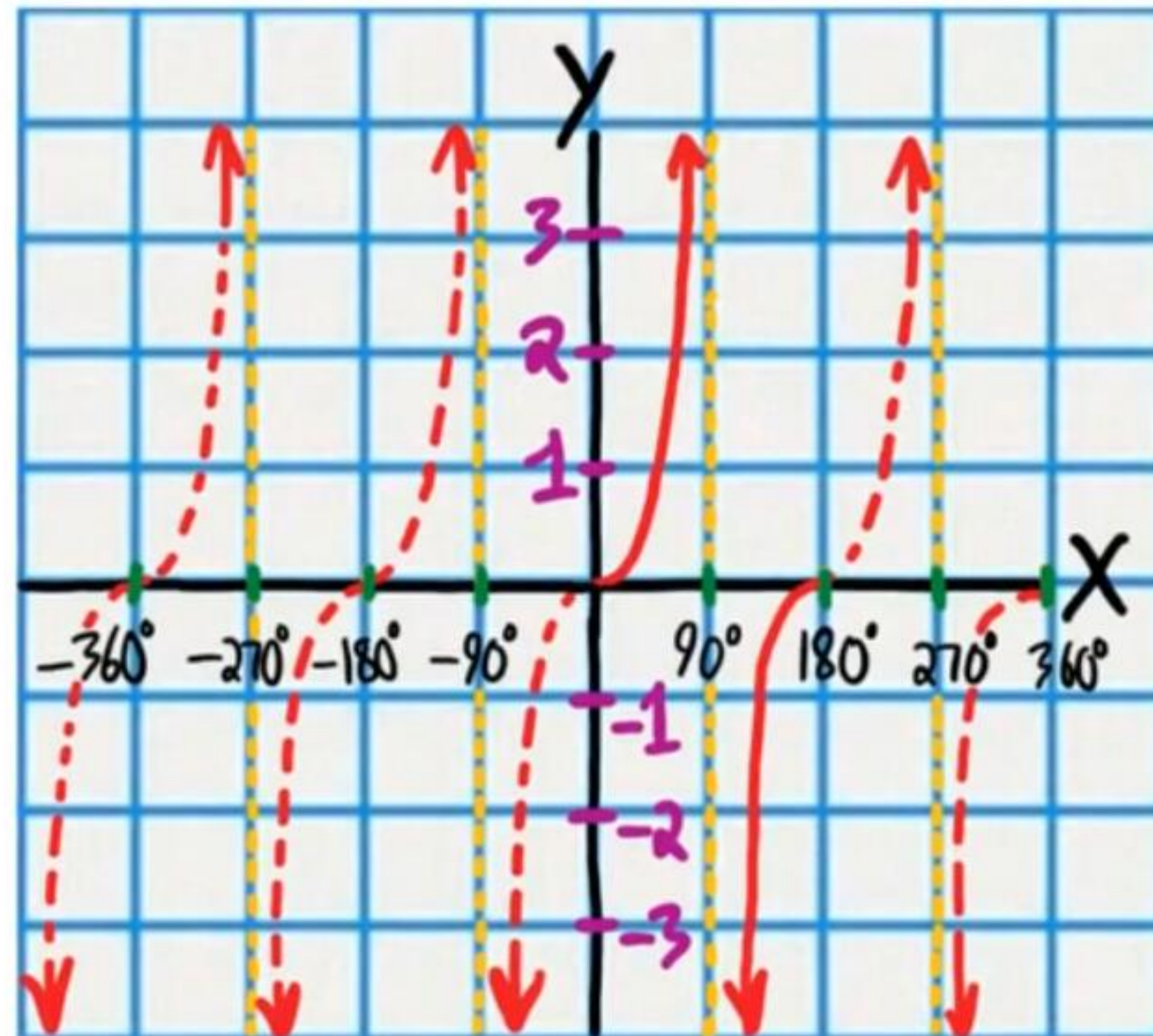
$$y = \sin x$$



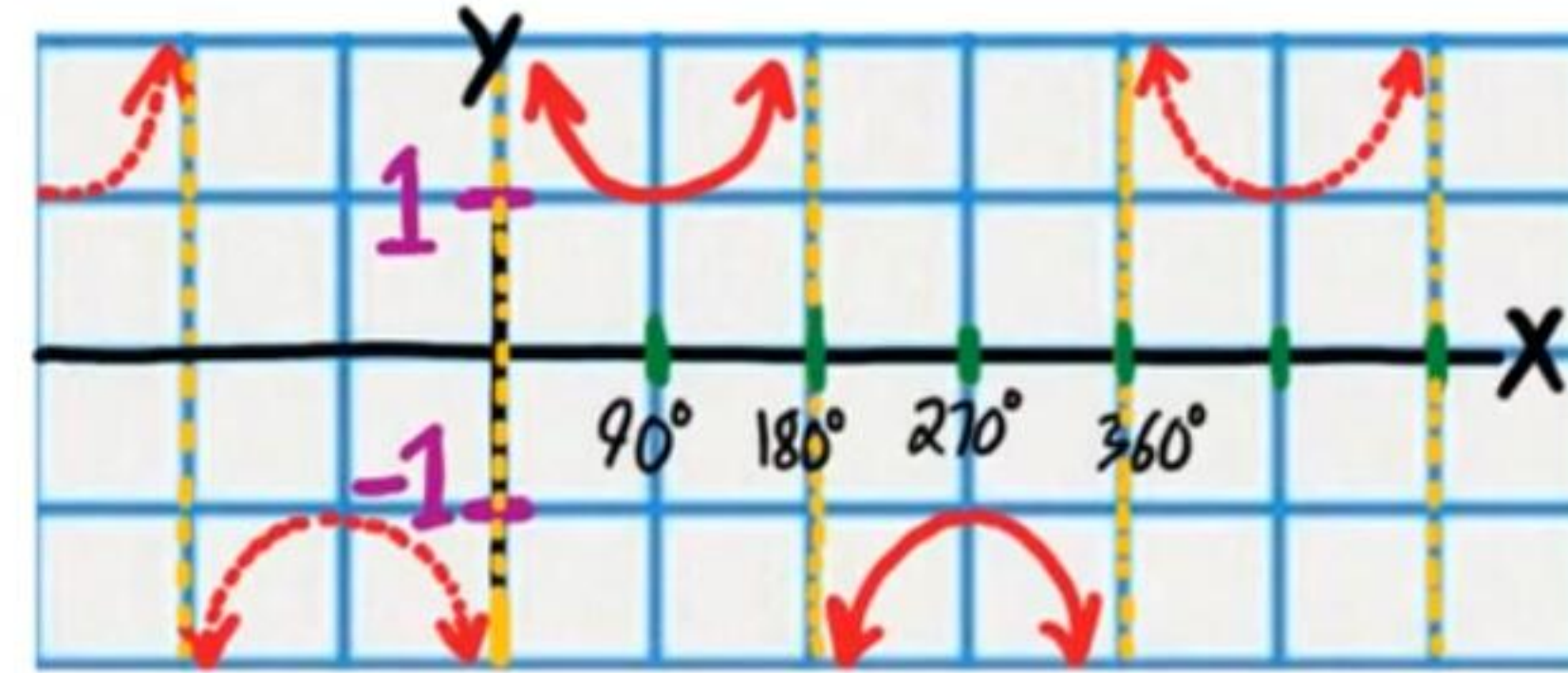
$$y = \cos x$$



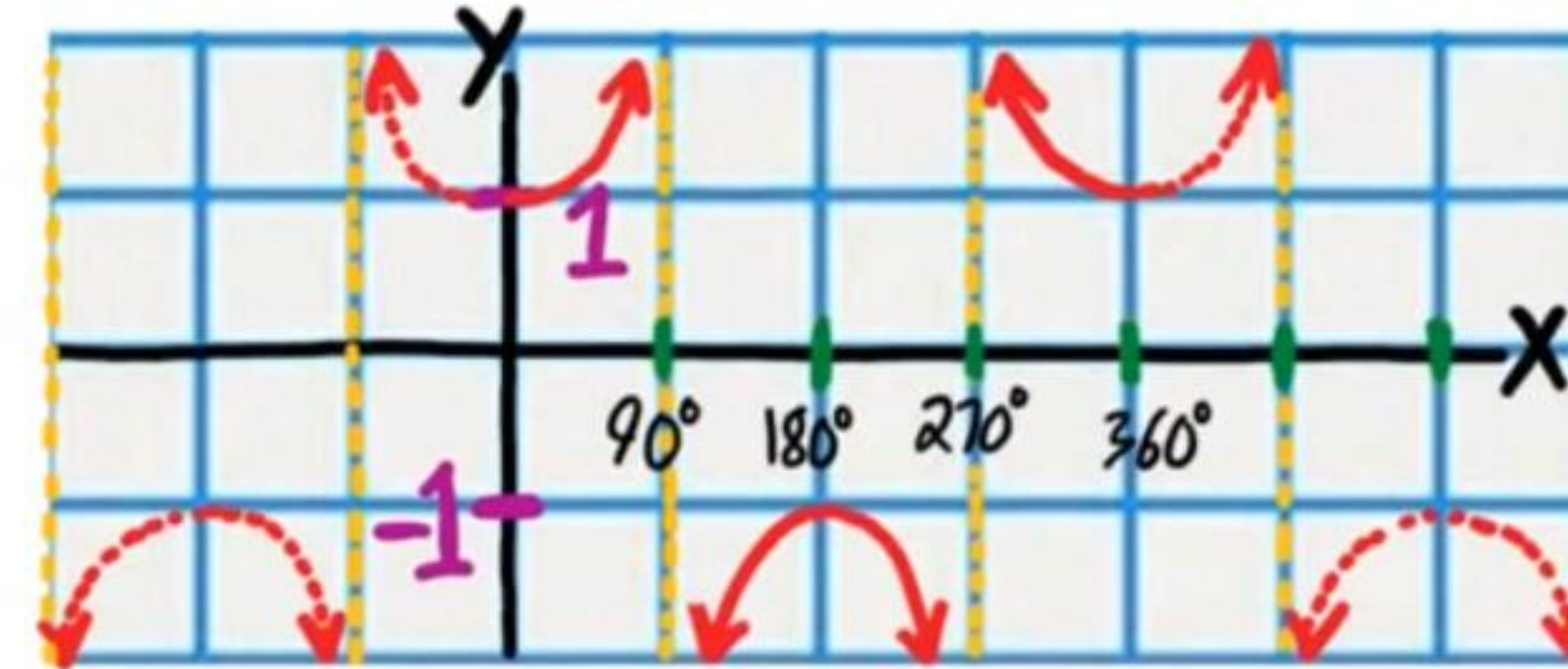
$$y = \tan x$$



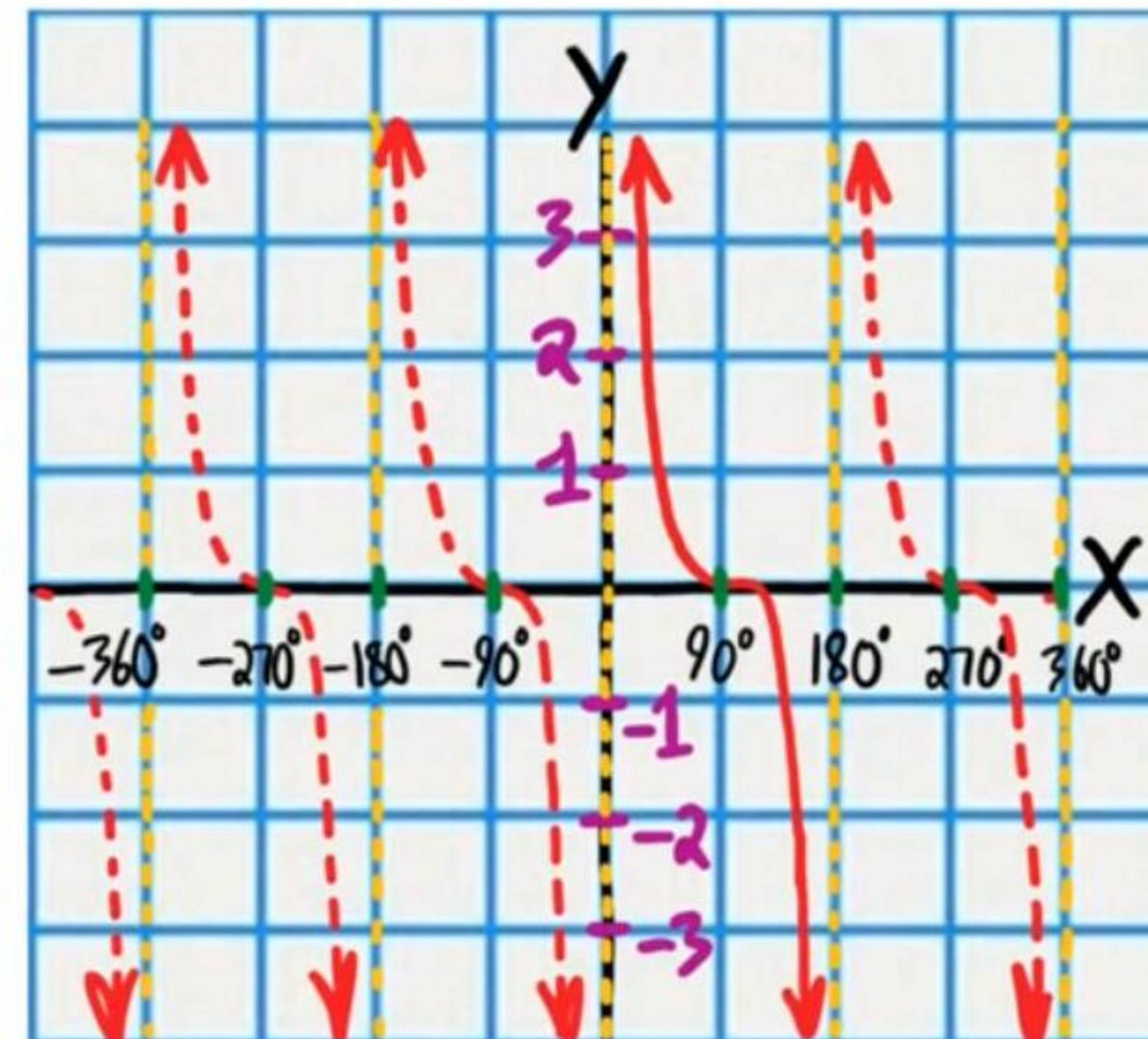
$$y = \csc x$$



$$y = \sec x$$



$$y = \cot x$$



(25) $\sec \pi$

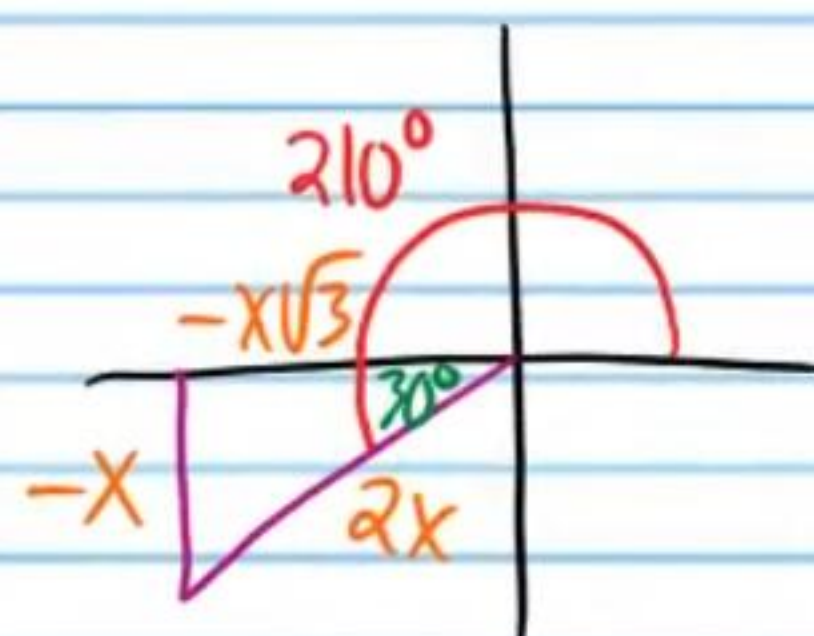
$$\pi \times \frac{180^\circ}{\pi} = \frac{180\pi^\circ}{\pi} = \boxed{180^\circ}$$

$$\sec \pi = \sec 180^\circ = \boxed{-1}$$

(26) $\tan \frac{7\pi}{6}$

$$\frac{7\pi}{6} \times \frac{180^\circ}{\pi} = \frac{1260\pi^\circ}{6\pi} = 210^\circ$$

$$\tan \frac{7\pi}{6} = \tan 210^\circ = \frac{-x}{-x\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$



(27) $\sin 2\pi$

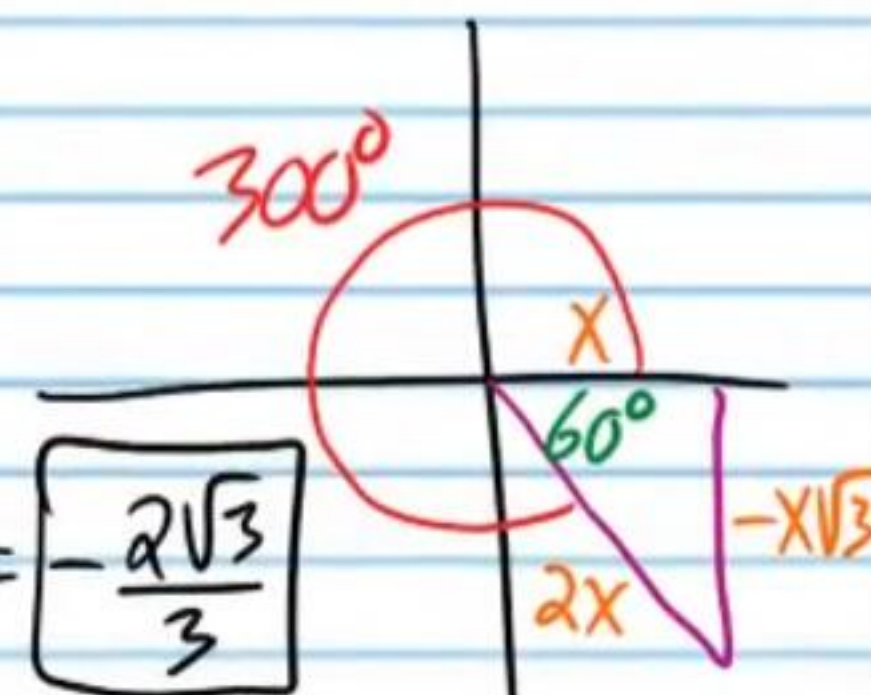
$$2\pi \times \frac{180^\circ}{\pi} = \frac{360\pi^\circ}{\pi} = 360^\circ$$

$$\sin 2\pi = \sin 360^\circ = \boxed{0}$$

(28) $\csc \frac{5\pi}{3}$

$$\frac{5\pi}{3} \times \frac{180^\circ}{\pi} = \frac{900\pi^\circ}{3\pi} = 300^\circ$$

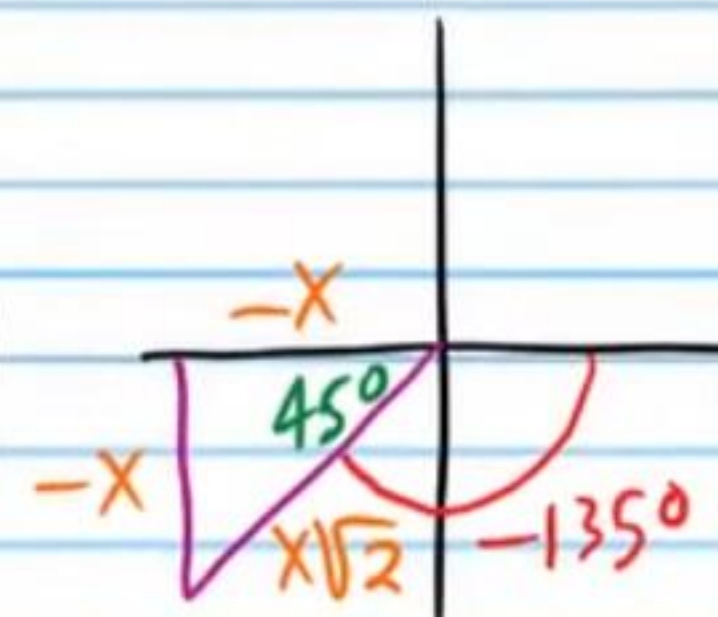
$$\csc \frac{5\pi}{3} = \csc 300^\circ = \frac{2x}{-x\sqrt{3}} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$$



(29) $\cot(-\frac{3\pi}{4})$

$$-\frac{3\pi}{4} \times \frac{180^\circ}{\pi} = -\frac{540\pi^\circ}{4\pi} = -135^\circ$$

$$\cot(-\frac{3\pi}{4}) = \cot(-135^\circ) = \frac{-x}{-x} = \boxed{1}$$



Find the value of each expression using a calculator.
Round to the nearest thousandth.

• (30) $\sin \frac{\pi}{9} \approx \boxed{0.342}$

(31) $\cos \frac{5\pi}{7} \approx \boxed{-0.623}$

(32) $\tan 3.4 \approx \boxed{0.264}$

• (33) $\sin \frac{2\pi}{5} \approx \boxed{0.951}$

(34) $\tan \frac{6\pi}{13} \approx \boxed{8.236}$

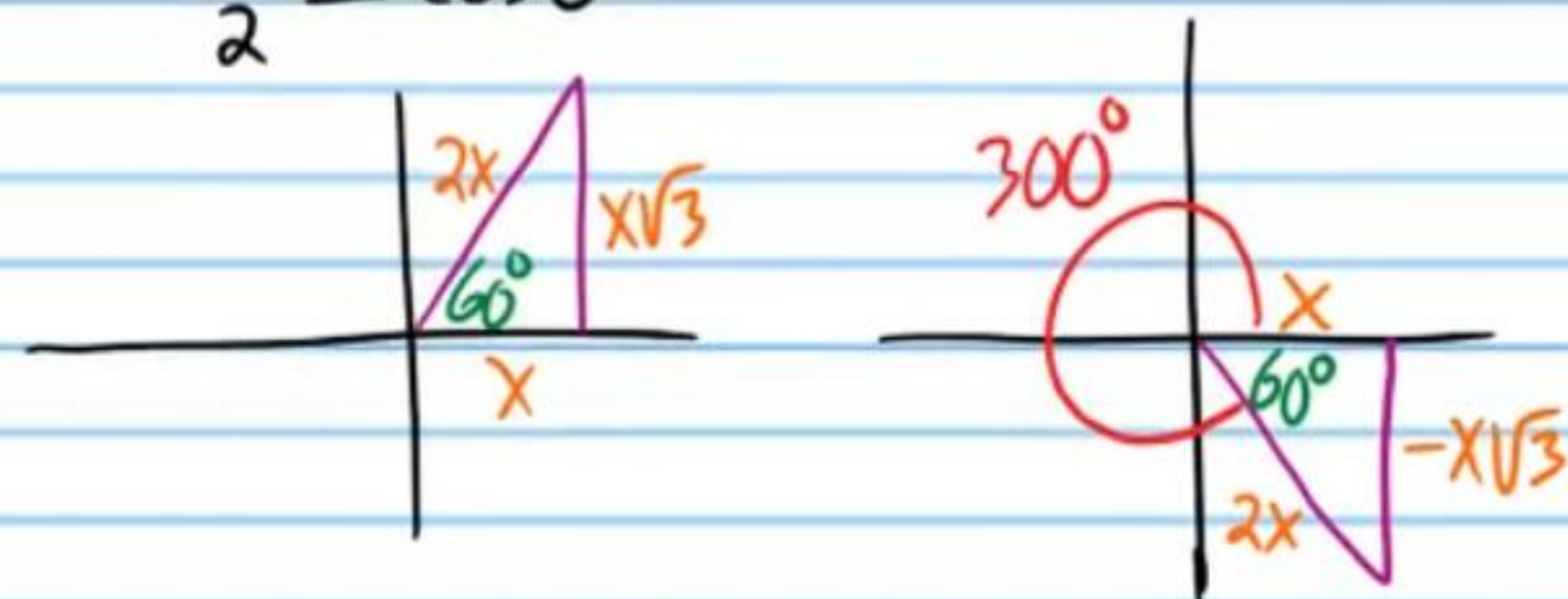
(35) $\cos 7 \approx \boxed{0.754}$

Find θ if $0 \leq \theta \leq 2\pi$.

• (36) $\cos \theta - 1 = -\cos \theta$

$$-1 = -2\cos \theta$$

$$\frac{1}{2} = \cos \theta$$



$$\theta = 60^\circ, 300^\circ$$

$$\theta = 60^\circ \times \frac{\pi}{180^\circ} = \frac{60\pi}{180} = \frac{\pi}{3}$$

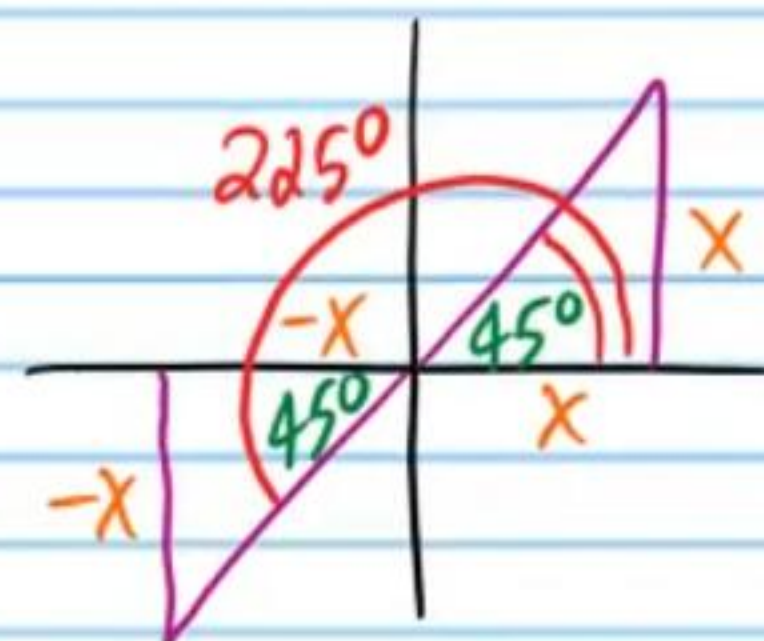
$$\theta = 300^\circ \times \frac{\pi}{180^\circ} = \frac{300\pi}{180} = \frac{5\pi}{3}$$

$$\boxed{\theta = \frac{\pi}{3}, \frac{5\pi}{3}}$$

(37) $\tan \theta + 7 \tan \theta = 8$

$$8 \tan \theta = 8$$

$$\tan \theta = 1$$



$$\theta = 45^\circ, 225^\circ$$

$$\theta = 45^\circ \times \frac{\pi}{180^\circ} = \frac{45\pi}{180} = \frac{\pi}{4}$$

$$\theta = 225^\circ \times \frac{\pi}{180^\circ} = \frac{225\pi}{180} = \frac{5\pi}{4}$$

$$\boxed{\theta = \frac{\pi}{4}, \frac{5\pi}{4}}$$

$$(38) -2\csc\theta + 4 = 5\csc\theta - 3$$

$$4 = 7\csc\theta - 3$$

$$7 = 7\csc\theta$$

$$1 = \csc\theta$$

$$\theta = 90^\circ$$

$$\theta = 90^\circ \times \frac{\pi}{180^\circ} = \frac{90\pi}{180} = \frac{\pi}{2}$$

$$\boxed{\theta = \frac{\pi}{2}}$$

$$(39) 6\sin\theta + 4 = 5\sin\theta + 3$$

$$\sin\theta = -1$$

$$\theta = 270^\circ$$

$$\theta = 270^\circ \times \frac{\pi}{180^\circ} = \frac{270\pi}{180}$$

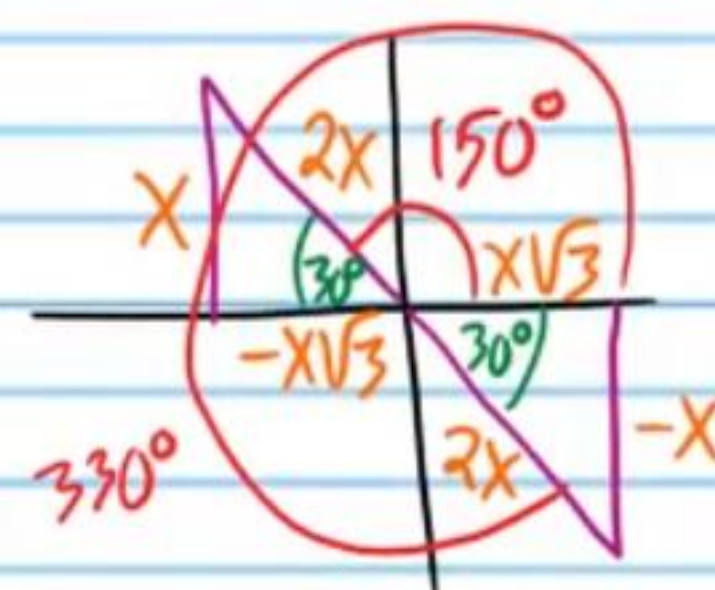
$$= \frac{3\pi}{2}$$

$$\boxed{\theta = \frac{3\pi}{2}}$$

$$(40) \cot\theta = -3\sqrt{3} - 2\cot\theta$$

$$3\cot\theta = -3\sqrt{3}$$

$$\cot\theta = -\sqrt{3}$$



$$\theta = 150^\circ, 330^\circ$$

$$\theta = 150^\circ \times \frac{\pi}{180^\circ} = \frac{150\pi}{180} = \frac{5\pi}{6}$$

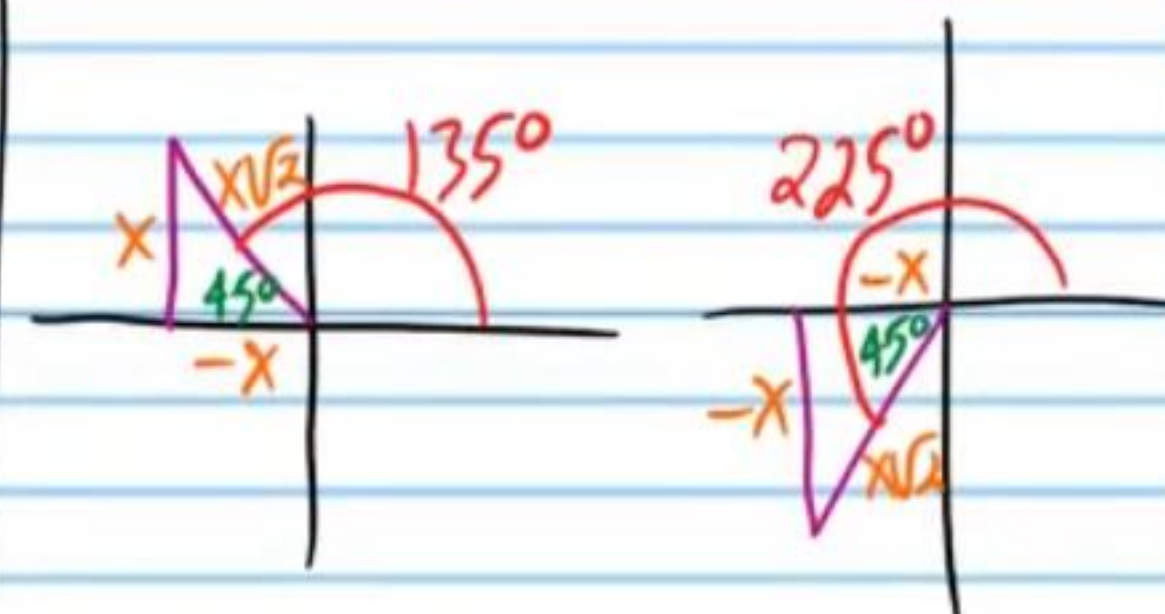
$$\theta = 330^\circ \times \frac{\pi}{180^\circ} = \frac{330\pi}{180} = \frac{11\pi}{6}$$

$$\boxed{\theta = \frac{5\pi}{6}, \frac{11\pi}{6}}$$

$$(41) 4\sec\theta + 2\sec\theta = -6\sqrt{2}$$

$$6\sec\theta = -6\sqrt{2}$$

$$\sec\theta = -\sqrt{2}$$



$$\theta = 135^\circ, 225^\circ$$

$$\theta = 135^\circ \times \frac{\pi}{180^\circ} = \frac{135\pi}{180}$$

$$= \frac{3\pi}{4}$$

$$\theta = 225^\circ \times \frac{\pi}{180^\circ} = \frac{225\pi}{180}$$

$$= \frac{5\pi}{4}$$

$$\boxed{\theta = \frac{3\pi}{4}, \frac{5\pi}{4}}$$